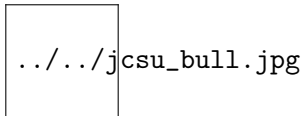


Ch. 10.1 Applications of Power Series

Calculus III | MTH 331 | Section A | Fall 2012

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Definition

A *power series* is an infinite series of the form

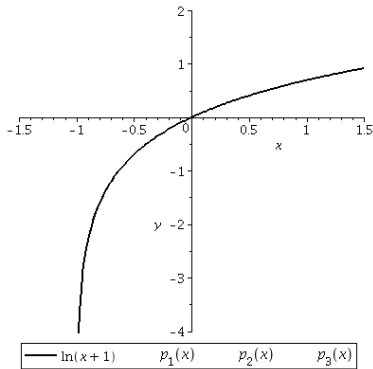
$$\sum_{k=0}^{\infty} c_k x^k = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + c_{n+1} x^{n+1} \dots$$

or, more generally,

$$\sum_{k=0}^{\infty} c_k (x - a)^k = c_0 + c_1 (x - a) + \cdots + c_n (x - a)^n + c_{n+1} (x - a)^{n+1} \dots$$

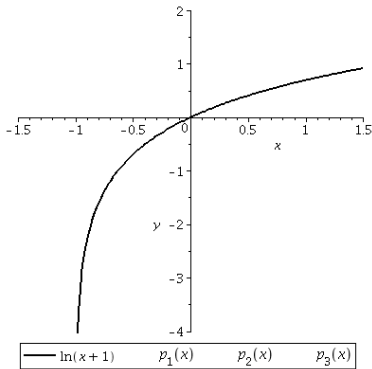
where each c_k and the **center** of the series a are constants.

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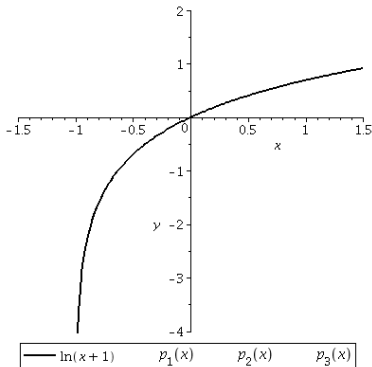
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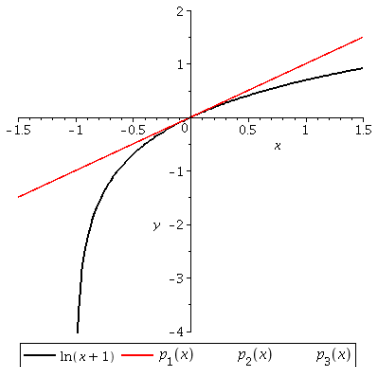


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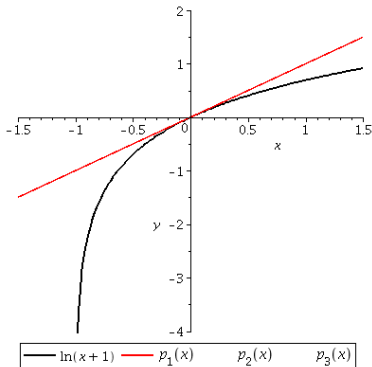
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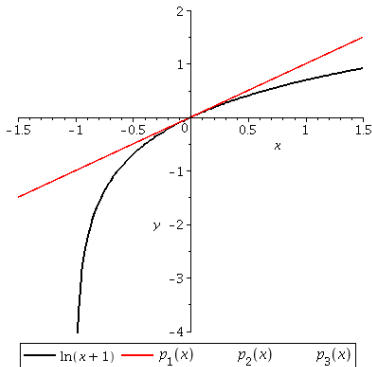
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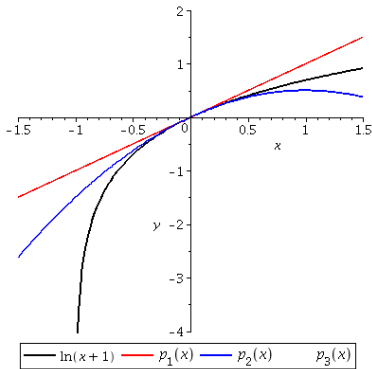
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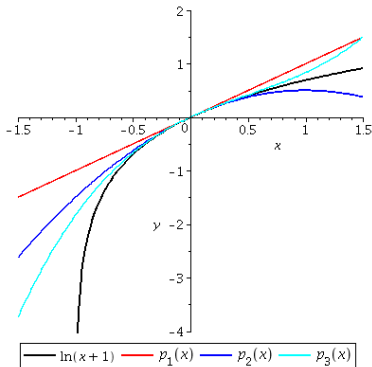
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Definition: Taylor Polynomials

Let f be a function with $f', f'', \dots, f^{(n)}$ defined at a . The n^{th} order Taylor polynomial for f with **center** a is,

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

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So $p_3(x) = 1 - \frac{1}{2} \frac{1}{(1)^{3/2}}(x) + \frac{3}{4} \frac{1}{(1)^{5/2}}(x)^2$, and

$$p_3(0.08) = 1 - \frac{1}{2}(0.08) + \frac{3}{4}(0.08)^2 \approx 1.02$$

Which is a decent approximation of $\sqrt{1.08} \approx 1.03923$.

Homework due 10/24

10.1: 7-8, 13-16, 21-22, 27-28, 33-36, 41-42